Nivedita's Blog

Structural Equation Modeling: A Practical Tutorial with R

What SEM is, why it matters, and how to fit a model using lavaan

R | STATISTICS | STRUCTURAL EQUATION MODELING | LATENT VARIABLES | LAVAAN | DATA SCIENCE

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1 Structural Equation Modeling: A Practical Tutorial with R

1.1 1. What is Structural Equation Modeling?

Structural Equation Modeling, usually called **SEM**, is a statistical framework used to study relationships among multiple variables at the same time.

In ordinary regression, we usually ask a question like:

> Does X predict Y?

For example:

> Does stress predict depression score?

SEM allows us to ask a broader and more realistic question:

> How do several observed and unobserved variables relate to each other simultaneously?

For example:

> Does socioeconomic status influence stress, and does stress then influence depression? Are stress and depression measured through several questionnaire items rather than one direct variable? Is the effect of socioeconomic status on depression partly mediated through stress?

SEM is useful because many scientific concepts are not directly observed. Variables such as intelligence, resilience, depression, socioeconomic status, anxiety, wellbeing, cognitive ability, and biological vulnerability are often **latent constructs**. We observe them indirectly through questionnaire items, test scores, biomarkers, or behavioral measures.

SEM combines two ideas:

1. **Measurement model** This describes how observed variables measure latent variables.

2. **Structural model** This describes how latent or observed variables are related to each other.

A simple way to remember SEM is:

> SEM = factor analysis + regression/path analysis

The `lavaan` package is one of the most widely used open-source R packages for SEM. It supports confirmatory factor analysis, SEM, latent growth models, and related latent variable models. (Journal of Statistical Software)

1.2 2. Why is SEM important?

SEM is important because many real-world research questions are not simple one-variable-to-one-variable problems.

In psychology, psychiatry, education, epidemiology, sociology, genetics, public health, and economics, researchers often work with complex systems. SEM helps model these systems in a structured way.

For example, suppose we are studying psychological resilience. We may believe that:

- family support improves resilience,
- resilience reduces depression,
- socioeconomic stress reduces resilience,
- depression is measured by several symptoms,
- resilience is measured by several questionnaire items.

A basic regression model may only test one part of this system. SEM allows us to test the full hypothesized structure.

SEM is especially useful when we want to:

- model latent variables,
- account for measurement error,
- test mediation,
- test direct and indirect effects,
- combine multiple regression equations,
- test whether a theoretical model fits the data,
- compare competing theoretical models.

This is why SEM is often used in fields where the theoretical structure matters as much as prediction.

1.3 3. Observed variables and latent variables

In SEM, variables are usually divided into two types.

1.3.1 Observed variables

Observed variables are directly measured in the dataset.

Examples:

- age
- sex
- income
- PHQ-9 item scores
- test scores
- biomarker values
- questionnaire responses

In path diagrams, observed variables are usually shown as rectangles.

1.3.2 Latent variables

Latent variables are not directly observed. They are inferred from multiple observed indicators.

Examples:

- depression
- anxiety
- resilience
- cognitive ability
- socioeconomic status
- inflammation
- wellbeing

In path diagrams, latent variables are usually shown as circles or ovals.

For example, depression may be measured using several questionnaire items:

```
depression =~ dep1 + dep2 + dep3 + dep4
```

Here, `depression` is a latent variable, and `dep1`, `dep2`, `dep3`, and `dep4` are observed indicators.

In `lavaan`, the operator `=~` defines a latent variable. The official lavaan SEM tutorial describes three core operators: `=~` for latent variable definitions, `~` for regressions, and `~~` for variances/covariances. (Lavaan)

1.4 4. The two parts of SEM

1.4.1 4.1 Measurement model

The measurement model asks:

> Are my observed variables good indicators of the latent construct?

For example:

```
depression =~ dep1 + dep2 + dep3 + dep4
resilience =~ res1 + res2 + res3 + res4
```

This says that depression is measured by four depression items, and resilience is measured by four resilience items.

This part of SEM is closely related to **Confirmatory Factor Analysis**, or CFA. CFA is used when we already have a hypothesis about which observed variables measure which latent variables. lavaan provides the `cfa()` function for this purpose. (Lavaan)

1.4.2 4.2 Structural model

The structural model asks:

> How are the latent variables related to each other?

For example:

```
depression ~ resilience + stress
resilience ~ social_support
```

depression ~ resilience + stress

resilience ~ social_support

This says:

- depression is predicted by resilience and stress,
- resilience is predicted by social support.

The structural part is like a system of regression models.

1.5 5. SEM syntax in lavaan

The basic syntax in `lavaan` is very readable.

1.5.1 Latent variable definition

```
latent_variable =~ item1 + item2 + item3
```

Example:

```
depression =~ dep1 + dep2 + dep3
```

1.5.2 Regression path

```
outcome ~ predictor
```

outcome ~ predictor

```
depression ~ stress + resilience
```

depression ~ stress + resilience

Example:

```
### Covariance

var1 ~~ var2
```

var1 ~ ~var2

Example:

```
stress ~~ social_support

### Variance

var1 ~~ var1

Example:
```

```
stress ~~ stress
```

```
stress ~ ~social_support
```

```
var1 ~ ~var1
```

```
Example:stress ~ ~stress
```

1.6 6. A simple SEM example

Let us create a simple example.

Suppose we are interested in three latent constructs:

1. **Stress**
2. **Resilience**
3. **Depression**

Each construct is measured by three observed questionnaire items.

We hypothesize that:

- stress increases depression,
- resilience reduces depression,
- stress reduces resilience,
- resilience partially mediates the effect of stress on depression.

The conceptual model is:

```
Stress  →  Resilience  →  Depression
   └───────────────────→  Depression
```

This means stress may affect depression directly and indirectly through resilience.

```
## 7. Complete R script using lavaan
```

```
# ============================================================
# Structural Equation Modeling Tutorial using lavaan
# ============================================================

# Install packages if needed
# install.packages("lavaan")
 install.packages("semPlot")
# install.packages("MASS")

library(lavaan)
library(semPlot)
library(MASS)

set.seed(123)

# ------------------------------------------------------------
# 1. Simulate synthetic data
# ------------------------------------------------------------

n <- 500

# Simulate latent variables
stress_latent <- rnorm(n, mean = 0, sd = 1)

resilience_latent <- -0.5 * stress_latent + rnorm(n, mean = 0, sd = (

depression_latent <- 0.6 * stress_latent - 0.7 * resilience_latent +
  rnorm(n, mean = 0, sd = 0.8)

# Create observed indicators with measurement error
data_sem <- data.frame(
  stress1 = 0.8 * stress_latent + rnorm(n, 0, 0.4),
  stress2 = 0.7 * stress_latent + rnorm(n, 0, 0.5),
  stress3 = 0.9 * stress_latent + rnorm(n, 0, 0.4),

  resil1 = 0.8 * resilience_latent + rnorm(n, 0, 0.4),
  resil2 = 0.7 * resilience_latent + rnorm(n, 0, 0.5),
  resil3 = 0.9 * resilience_latent + rnorm(n, 0, 0.4),

  dep1 = 0.8 * depression_latent + rnorm(n, 0, 0.4),
  dep2 = 0.7 * depression_latent + rnorm(n, 0, 0.5),
  dep3 = 0.9 * depression_latent + rnorm(n, 0, 0.4)
)

head(data_sem)
summary(data_sem)
```

```
The downloaded binary packages are in

/var/folders/vm/xtvv8jb542s04t61c12lsnvr0000gn/T//RtmpWIXWUT/download
ed_packages
```

A data.frame: 6 × 9

	stress1	stress2	stress3	resil1	resil2	resil3	dep1
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	-0.7767752	-0.6481348	-0.77595113	-0.22114415	0.5982737	-0.102528898	-0.4598761
2	-0.3070449	-0.0426553	0.02256535	-0.67499896	-1.1793024	-0.351837840	-0.6746907
3	0.8861274	0.8203012	1.12103167	-0.54560701	-0.9125343	0.306268174	1.2368075
4	0.3072342	0.6589697	-0.15013607	0.17356204	0.2572333	-0.004627744	-0.7595522
5	0.5515722	0.1775694	0.42611280	0.02181441	-0.6751701	-1.955223221	-0.5580200
6	2.2229374	0.8929113	1.35330993	-0.76188637	-0.7179487	0.041845017	2.5151611

```
    stress1             stress2              stress3             resil1
 Min.   :-2.68041   Min.   :-2.691356   Min.   :-2.64489   Min.
:-2.82284
 1st Qu.:-0.54070   1st Qu.:-0.543447   1st Qu.:-0.53635   1st
Qu.:-0.64442
 Median : 0.06657   Median :-0.029942   Median : 0.04875   Median :
0.01097
 Mean   : 0.05116   Mean   :-0.007807   Mean   : 0.04066   Mean
:-0.02437
 3rd Qu.: 0.57855   3rd Qu.: 0.524629   3rd Qu.: 0.61250   3rd Qu.:
0.53320
 Max.   : 2.78049   Max.   : 2.680151   Max.   : 3.15869   Max.   :
2.26858
     resil2             resil3              dep1                dep2
 Min.   :-2.42576   Min.   :-3.19028   Min.   :-3.012669   Min.
:-2.59815
 1st Qu.:-0.59795   1st Qu.:-0.66951   1st Qu.:-0.761669   1st
Qu.:-0.62917
 Median : 0.04993   Median :-0.02788   Median : 0.006298   Median :
0.01076
 Mean   :-0.01127   Mean   :-0.03598   Mean   : 0.037102   Mean   :
0.04762
 3rd Qu.: 0.52862   3rd Qu.: 0.59763   3rd Qu.: 0.785262   3rd Qu.:
0.76276
 Max.   : 2.48716   Max.   : 2.97850   Max.   : 3.457041   Max.   :
3.43147
```

```
      dep3
 Min.   :-3.66353
 1st Qu.:-0.84296
 Median : 0.07153
 Mean   : 0.06997
 3rd Qu.: 0.97918
 Max.   : 4.09965
```

1.7 8. Define the SEM model

```
# ------------------------------------------------------------
# 2. Define SEM model
# ------------------------------------------------------------

model_sem <- '

  # Measurement model
  stress =~ stress1 + stress2 + stress3
  resilience =~ resil1 + resil2 + resil3
  depression =~ dep1 + dep2 + dep3

  # Structural model
  resilience ~ a * stress
  depression ~ b * resilience + c * stress

  # Indirect, direct, and total effects
  indirect := a * b
  direct := c
  total := c + (a * b)

'
```

In this model:

```
stress =~ stress1 + stress2 + stress3
```

means that the latent variable `stress` is measured by three observed items.

```
resilience ~ a * stress
```

```
resilience ~ a * stress
```

means that stress predicts resilience. The path coefficient is labeled `a`.

```
depression ~ b * resilience + c * stress
```


```
depression ~ b * resilience + c * stress
```


means that depression is predicted by both resilience and stress. The path from resilience to depression is labeled `b`, and the direct path from stress to depression is labeled `c`.

The indirect effect is calculated as:

indirect := a * b

This represents:

1.8 9. Fit the SEM model

```
stress → resilience → depression
```

```
# -----------------------------------------------------------
# 3. Fit the model
# -----------------------------------------------------------

fit_sem <- sem(
  model_sem,
  data = data_sem,
  meanstructure = TRUE
)

summary(
  fit_sem,
  fit.measures = TRUE,
  standardized = TRUE,
  rsquare = TRUE
)
```

$header

$lavaan.version

'0.7-2'

$sam.approach

FALSE

$optim.method

'nlminb'

$optim.iterations

34

$optim.converged

TRUE

$optim

$estimator

'ML'

$estimator.args

$optim.method

'nlminb'

$npar

30

$eq.constraints

FALSE

$nrow.ceq.jac

0

$nrow.cin.jac

0

$nrow.con.jac

0

$con.jac.rank

0

$data

$ngroups

1

$nobs

500

$test

$standard =

$test

'standard'

$stat

39.5933180430763

$stat.group

39.5933180430763

$df

24

$refdistr

'chisq'

$pvalue

0.023638187618621

$fit

npar: 30 **fmin:** 0.0395933180430763 **chisq:** 39.5933180430763 **df:** 24 **pvalue:** 0.023638187618621 **baseline.chisq:** 4010.69045035666 **baseline.df:** 36 **baseline.pvalue:** 0 **cfi:** 0.996076847181476 **tli:** 0.994115270772214 **logl:** -4261.75022147873 **unrestricted.logl:** -4241.95356245719 **aic:** 8583.50044295747 **bic:** 8709.93868591013 **ntotal:** 500 **bic2:** 8614.71683163778 **rmsea:** 0.0360477900883863 **rmsea.ci.lower:** 0.0133206537558792 **rmsea.ci.upper:** 0.0554785467010791 **rmsea.ci.level:** 0.9 **rmsea.pvalue:** 0.872609329442447 **rmsea.close.h0:** 0.05 **rmsea.notclose.pvalue:** 2.92071859358351e-05 **rmsea.notclose.h0:** 0.08 **srmr:** 0.0195584346342506 **gfi:** 0.994230146309336 **gfi.ci.lower:** 0.985281597136529 **gfi.ci.upper:** 0.999811174122434 **gfi.ci.level:** 0.9

$pe

A data.frame: 47 × 11

	lhs	op	rhs	label	exo	est	se	z	p
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<
1	stress	=~	stress1		0	1.000000000	0.00000000	NA	N
2	stress	=~	stress2		0	0.888712566	0.03965218	22.4127048	0.
3	stress	=~	stress3		0	1.165333138	0.04332610	26.8967905	0.
4	resilience	=~	resil1		0	1.000000000	0.00000000	NA	N
5	resilience	=~	resil2		0	0.851920295	0.03685346	23.1164239	0.
6	resilience	=~	resil3		0	1.123935553	0.04011596	28.0171692	0.
7	depression	=~	dep1		0	1.000000000	0.00000000	NA	N
8	depression	=~	dep2		0	0.852686146	0.02485828	34.3018961	0.
9	depression	=~	dep3		0	1.146769075	0.02726355	42.0623568	0.
10	resilience	~	stress	a	0	-0.536162652	0.04798759	-11.1729444	0.
11	depression	~	resilience	b	0	-0.638717896	0.05641726	-11.3213206	0.
12	depression	~	stress	c	0	0.727534488	0.05968082	12.1904241	0.
13	stress1	~~	stress1		0	0.179796776	0.01664951	10.7989246	0.
14	stress2	~~	stress2		0	0.233500062	0.01801124	12.9641268	0.
15	stress3	~~	stress3		0	0.140384235	0.01830990	7.6671204	1.
16	resil1	~~	resil1		0	0.145902412	0.01563347	9.3326929	0.
17	resil2	~~	resil2		0	0.237165208	0.01806223	13.1304497	0.
18	resil3	~~	resil3		0	0.164328886	0.01895948	8.6673739	0.
19	dep1	~~	dep1		0	0.165647777	0.01695952	9.7672442	0.
20	dep2	~~	dep2		0	0.232658848	0.01803935	12.8972957	0.
21	dep3	~~	dep3		0	0.173267305	0.02052209	8.4429653	0.
22	stress	~~	stress		0	0.552502385	0.04651494	11.8779564	0.
23	resilience	~~	resilience		0	0.436318467	0.03668041	11.8951349	0.
24	depression	~~	depression		0	0.433759482	0.03669715	11.8199764	0.
25	stress1	~1			0	0.051164260	0.03827007	1.3369262	1.
26	stress2	~1			0	-0.007807425	0.03660251	-0.2133030	8.
27	stress3	~1			0	0.040657971	0.04220624	0.9633167	3.
28	resil1	~1			0	-0.024374481	0.03849802	-0.6331359	5.
29	resil2	~1			0	-0.011269369	0.03658152	-0.3080618	7.

	lhs	op	rhs	label	exo	est	se	z	p
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<
30	resil3	~1			0	-0.035980507	0.04280506	-0.8405667	4.
31	dep1	~1			0	0.037102437	0.05310287	0.6986898	4.
32	dep2	~1			0	0.047620784	0.04769410	0.9984627	3.
33	dep3	~1			0	0.069974908	0.06016034	1.1631401	2.
37	indirect	:=	a*b	indirect	0	0.342456681	0.04061961	8.4308214	0.
38	direct	:=	c	direct	0	0.727534488	0.05968082	12.1904241	0.
39	total	:=	c+(a*b)	total	0	1.069991169	0.06169700	17.3426763	0.
40	stress1	r2	stress1		0	0.754476332	NA	NA	N
41	stress2	r2	stress2		0	0.651425844	NA	NA	N
42	stress3	r2	stress3		0	0.842385896	NA	NA	N
43	resil1	r2	resil1		0	0.803113666	NA	NA	N
44	resil2	r2	resil2		0	0.645547872	NA	NA	N
45	resil3	r2	resil3		0	0.820628388	NA	NA	N
46	dep1	r2	dep1		0	0.882515768	NA	NA	N
47	dep2	r2	dep2		0	0.795440236	NA	NA	N
48	dep3	r2	dep3		0	0.904252809	NA	NA	N
49	resilience	r2	resilience		0	0.266872214	NA	NA	N
50	depression	r2	depression		0	0.651405541	NA	NA	N

The `sem()` function fits the structural equation model. The argument `standardized = TRUE` gives standardized estimates, which are often easier to interpret.

1.9 10. Understanding the output

The output contains several important parts.

1.9.1 10.1 Factor loadings

Factor loadings show how strongly each observed item measures its latent construct.

Example:

```
stress =~
  stress1
  stress2
  stress3
```

If the standardized loading of `stress1` is 0.85, it means `stress1` is a strong indicator of the latent stress factor.

A rough interpretation:

Standardized loading	Interpretation
0.30	weak
0.50	moderate
0.70 or higher	strong

These are only rough guidelines. Interpretation depends on the field and measurement instrument.

1.9.2 10.2 Regression paths

The structural paths tell us how latent variables predict each other.

Example:

```
resilience ~ stress
depression ~ resilience + stress
```

If the standardized coefficient from stress to resilience is negative, it means higher stress is associated with lower resilience.

If the coefficient from resilience to depression is negative, it means higher resilience is associated with lower depression.

If the coefficient from stress to depression is positive, it means higher stress is associated with higher depression.

1.9.3 10.3 R-squared values

R-squared tells us how much variance is explained in each endogenous variable.

For example:

```
R2 for resilience = 0.25
R2 for depression = 0.60
```

This means:

- 25% of the variance in resilience is explained by stress,
- 60% of the variance in depression is explained by stress and resilience.

1.10 11. Model fit indices

SEM does not only estimate coefficients. It also asks:

> Does the proposed model reproduce the observed covariance structure reasonably well?

This is one of the key differences between SEM and ordinary regression.

Common fit indices include:

1.10.1 Chi-square test

The chi-square test evaluates exact model fit.

A significant chi-square suggests that the model-implied covariance matrix differs from the observed covariance matrix.

However, chi-square is sensitive to sample size. With large samples, even small misfit can become statistically significant.

1.10.2 CFI: Comparative Fit Index

CFI compares the proposed model with a baseline model.

Common rough guidelines:

- CFI > 0.90: acceptable
- CFI > 0.95: good

1.10.3 TLI: Tucker-Lewis Index

TLI is similar to CFI but penalizes model complexity.

Common rough guidelines:

- TLI > 0.90: acceptable
- TLI > 0.95: good

1.10.4 RMSEA: Root Mean Square Error of Approximation

RMSEA measures approximate fit.

Common rough guidelines:

- RMSEA < 0.08: acceptable
- RMSEA < 0.05: good

1.10.5 SRMR: Standardized Root Mean Square Residual

SRMR measures the average standardized difference between observed and predicted correlations.

Common rough guidelines:

- SRMR < 0.08: acceptable

These cutoffs should not be used mechanically. They are guidelines, not universal rules.

You can extract fit measures in R:

```
fitMeasures(
  fit_sem,
  c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "srmr")
)
```

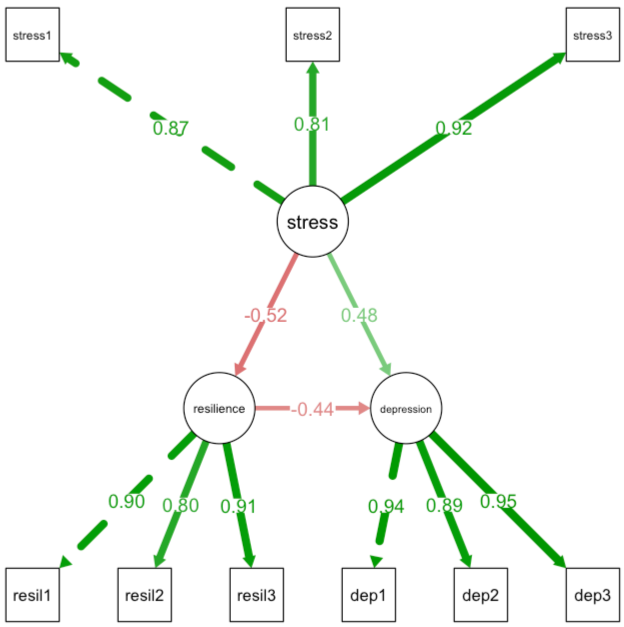
chisq: 39.5933180430763 **df:** 24 **pvalue:** 0.023638187618621 **cfi:** 0.996076847181476 **tli:** 0.994115270772214 **rmsea:** 0.0360477900883863 **srmr:** 0.0195584346342506

1.11 12. Visualize the SEM model

```
# ------------------------------------------------------------
# 4. Plot the SEM model
# ------------------------------------------------------------

semPaths(
  fit_sem,
```

```
  what = "std",
  layout = "tree",
  edge.label.cex = 1.0,
  sizeMan = 6,
  sizeLat = 8,
  residuals = FALSE,
  intercepts = FALSE,
  nCharNodes = 0
)
```



This creates a path diagram.

In the diagram:

- circles represent latent variables,
- rectangles represent observed variables,
- arrows represent regression or measurement paths,
- numbers on arrows are usually standardized coefficients.

1.12 13. Mediation in SEM

One of the most useful applications of SEM is mediation analysis.

In our example:

```
stress → resilience → depression
```

This means stress may affect depression indirectly through resilience.

The indirect effect is:

indirect := a * b

Where:

- `a` = effect of stress on resilience,
- `b` = effect of resilience on depression.

The total effect is:

total := direct + indirect

In lavaan:

```
parameterEstimates(
  fit_sem,
  standardized = TRUE
)
```

A lavaan.data.frame: 39 × 12

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
stress	=~	stress1		1.000000000	0.00000000	NA	NA
stress	=~	stress2		0.888712566	0.03965218	22.4127048	0.000000e+
stress	=~	stress3		1.165333138	0.04332610	26.8967905	0.000000e+
resilience	=~	resil1		1.000000000	0.00000000	NA	NA
resilience	=~	resil2		0.851920295	0.03685346	23.1164239	0.000000e+
resilience	=~	resil3		1.123935553	0.04011596	28.0171692	0.000000e+
depression	=~	dep1		1.000000000	0.00000000	NA	NA
depression	=~	dep2		0.852686146	0.02485828	34.3018961	0.000000e+
depression	=~	dep3		1.146769075	0.02726355	42.0623568	0.000000e+
resilience	~	stress	a	-0.536162652	0.04798759	-11.1729444	0.000000e+

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
depression	~	resilience	b	-0.638717896	0.05641726	-11.3213206	0.000000e+
depression	~	stress	c	0.727534488	0.05968082	12.1904241	0.000000e+
stress1	~~	stress1		0.179796776	0.01664951	10.7989246	0.000000e+
stress2	~~	stress2		0.233500062	0.01801124	12.9641268	0.000000e+
stress3	~~	stress3		0.140384235	0.01830990	7.6671204	1.754152e-
resil1	~~	resil1		0.145902412	0.01563347	9.3326929	0.000000e+
resil2	~~	resil2		0.237165208	0.01806223	13.1304497	0.000000e+
resil3	~~	resil3		0.164328886	0.01895948	8.6673739	0.000000e+
dep1	~~	dep1		0.165647777	0.01695952	9.7672442	0.000000e+
dep2	~~	dep2		0.232658848	0.01803935	12.8972957	0.000000e+
dep3	~~	dep3		0.173267305	0.02052209	8.4429653	0.000000e+
stress	~~	stress		0.552502385	0.04651494	11.8779564	0.000000e+
resilience	~~	resilience		0.436318467	0.03668041	11.8951349	0.000000e+
depression	~~	depression		0.433759482	0.03669715	11.8199764	0.000000e+
stress1	~1			0.051164260	0.03827007	1.3369262	1.812467e-
stress2	~1			-0.007807425	0.03660251	-0.2133030	8.310907e-
stress3	~1			0.040657971	0.04220624	0.9633167	3.353886e-
resil1	~1			-0.024374481	0.03849802	-0.6331359	5.266449e-
resil2	~1			-0.011269369	0.03658152	-0.3080618	7.580353e-
resil3	~1			-0.035980507	0.04280506	-0.8405667	4.005907e-
dep1	~1			0.037102437	0.05310287	0.6986898	4.847459e-
dep2	~1			0.047620784	0.04769410	0.9984627	3.180550e-
dep3	~1			0.069974908	0.06016034	1.1631401	2.447726e-
stress	~1			0.000000000	0.00000000	NA	NA
resilience	~1			0.000000000	0.00000000	NA	NA
depression	~1			0.000000000	0.00000000	NA	NA
indirect	:=	a*b	indirect	0.342456681	0.04061961	8.4308214	0.000000e+
direct	:=	c	direct	0.727534488	0.05968082	12.1904241	0.000000e+
total	:=	c+(a*b)	total	1.069991169	0.06169700	17.3426763	0.000000e+

This will show the direct, indirect, and total effects.

1.13 14. Bootstrapped confidence intervals for indirect effects

Indirect effects often have non-normal sampling distributions. Bootstrapping is commonly used for mediation.

```
fit_sem_boot <- sem(
  model_sem,
  data = data_sem,
  se = "bootstrap",
  bootstrap = 1000,
  meanstructure = TRUE
)

parameterEstimates(
  fit_sem_boot,
  boot.ci.type = "perc",
  standardized = TRUE
)
```

A lavaan.data.frame: 39 × 12

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
stress	=~	stress1		1.000000000	0.00000000	NA	NA
stress	=~	stress2		0.888712566	0.03875278	22.9328715	0.000000e+
stress	=~	stress3		1.165333138	0.04330515	26.9098040	0.000000e+
resilience	=~	resil1		1.000000000	0.00000000	NA	NA
resilience	=~	resil2		0.851920295	0.03808626	22.3681820	0.000000e+
resilience	=~	resil3		1.123935553	0.04317346	26.0330200	0.000000e+
depression	=~	dep1		1.000000000	0.00000000	NA	NA
depression	=~	dep2		0.852686146	0.02355667	36.1972280	0.000000e+
depression	=~	dep3		1.146769075	0.02873547	39.9077899	0.000000e+
resilience	~	stress	a	-0.536162652	0.04772853	-11.2335886	0.000000e+
depression	~	resilience	b	-0.638717896	0.05169940	-12.3544541	0.000000e+

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
depression	~	stress	c	0.727534488	0.05869613	12.3949303	0.000000e+
stress1	~~	stress1		0.179796776	0.01502245	11.9685414	0.000000e+
stress2	~~	stress2		0.233500062	0.01885213	12.3858737	0.000000e+
stress3	~~	stress3		0.140384235	0.01843239	7.6161708	2.620126e-
resil1	~~	resil1		0.145902412	0.01595743	9.1432259	0.000000e+
resil2	~~	resil2		0.237165208	0.01709337	13.8746937	0.000000e+
resil3	~~	resil3		0.164328886	0.02084598	7.8830019	3.108624e-
dep1	~~	dep1		0.165647777	0.01906716	8.6875974	0.000000e+
dep2	~~	dep2		0.232658848	0.01764236	13.1875122	0.000000e+
dep3	~~	dep3		0.173267305	0.02048533	8.4581171	0.000000e+
stress	~~	stress		0.552502385	0.04695879	11.7656873	0.000000e+
resilience	~~	resilience		0.436318467	0.03765419	11.5875151	0.000000e+
depression	~~	depression		0.433759482	0.03872977	11.1996388	0.000000e+
stress1	~1			0.051164260	0.03771417	1.3566322	1.748981e-
stress2	~1			-0.007807425	0.03691312	-0.2115081	8.324908e-
stress3	~1			0.040657971	0.04279942	0.9499654	3.421298e-
resil1	~1			-0.024374481	0.03766567	-0.6471272	5.175496e-
resil2	~1			-0.011269369	0.03674697	-0.3066748	7.590909e-
resil3	~1			-0.035980507	0.04224272	-0.8517564	3.943493e-
dep1	~1			0.037102437	0.05255557	0.7059658	4.802094e-
dep2	~1			0.047620784	0.04743160	1.0039886	3.153841e-
dep3	~1			0.069974908	0.05921142	1.1817807	2.372927e-
stress	~1			0.000000000	0.00000000	NA	NA
resilience	~1			0.000000000	0.00000000	NA	NA
depression	~1			0.000000000	0.00000000	NA	NA
indirect	:=	a*b	indirect	0.342456681	0.03929570	8.7148636	0.000000e+
direct	:=	c	direct	0.727534488	0.05869613	12.3949303	0.000000e+
total	:=	c+(a*b)	total	1.069991169	0.06601319	16.2087480	0.000000e+

This gives bootstrap confidence intervals for the indirect effect.

If the confidence interval for the indirect effect does not include zero, that supports evidence for mediation.

1.14 15. CFA before SEM

In practice, it is often useful to first fit the measurement model alone.

```
# ------------------------------------------------------------
# 5. Confirmatory Factor Analysis first
# ------------------------------------------------------------

model_cfa <- '

  stress =~ stress1 + stress2 + stress3
  resilience =~ resil1 + resil2 + resil3
  depression =~ dep1 + dep2 + dep3

'

fit_cfa <- cfa(
  model_cfa,
  data = data_sem,
  meanstructure = TRUE
)

summary(
  fit_cfa,
  fit.measures = TRUE,
  standardized = TRUE
)
```

$header

$lavaan.version

'0.7-2'

$sam.approach

FALSE

$optim.method

'nlminb'

$optim.iterations

43

$optim.converged

TRUE

$optim

$estimator

'ML'

$estimator.args

$optim.method

'nlminb'

$npar

30

$eq.constraints

FALSE

$nrow.ceq.jac

0

$nrow.cin.jac

0

$nrow.con.jac

0

$con.jac.rank

0

$data

$ngroups

1

$nobs

500

$test

$standard =

$test

'standard'

$stat

39.5933180434271

$stat.group

39.5933180434271

$df

24

$refdistr

'chisq'

$pvalue

0.0236381876165893

$fit

npar: 30 **fmin:** 0.0395933180434271 **chisq:** 39.5933180434271 **df:** 24 **pvalue:** 0.0236381876165893 **baseline.chisq:** 4010.69045035666 **baseline.df:** 36 **baseline.pvalue:** 0 **cfi:** 0.996076847181388 **tli:** 0.994115270772082 **logl:** -4261.75022147891 **unrestricted.logl:** -4241.95356245719 **aic:** 8583.50044295782 **bic:** 8709.93868591049 **ntotal:** 500 **bic2:** 8614.71683163813 **rmsea:** 0.0360477900887918 **rmsea.ci.lower:** 0.0133206537566257 **rmsea.ci.upper:** 0.0554785467014227 **rmsea.ci.level:** 0.9 **rmsea.pvalue:** 0.872609329435695 **rmsea.close.h0:** 0.05 **rmsea.notclose.pvalue:** 2.92071859397014e-05 **rmsea.notclose.h0:** 0.08 **srmr:** 0.0195584666176241 **gfi:** 0.994230150253581 **gfi.ci.lower:** 0.98528160224032 **gfi.ci.upper:** 0.999811176769756 **gfi.ci.level:** 0.9

$pe

A data.frame: 33 × 10

	lhs	op	rhs	exo	est	se	z	pvalue
	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	stress	=~	stress1	0	1.000000000	0.00000000	NA	NA
2	stress	=~	stress2	0	0.888712491	0.03965218	22.4127035	0.000000e
3	stress	=~	stress3	0	1.165332895	0.04332610	26.8967848	0.000000e
4	resilience	=~	resil1	0	1.000000000	0.00000000	NA	NA
5	resilience	=~	resil2	0	0.851920122	0.03685347	23.1164167	0.000000e
6	resilience	=~	resil3	0	1.123935775	0.04011596	28.0171702	0.000000e

	lhs	op	rhs	exo	est	se	z	pvalue
	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
7	depression	=~	dep1	0	1.000000000	0.00000000	NA	NA
8	depression	=~	dep2	0	0.852686291	0.02485828	34.3018972	0.000000e
9	depression	=~	dep3	0	1.146769204	0.02726356	42.0623465	0.000000e
10	stress1	~~	stress1	0	0.179796686	0.01664951	10.7989206	0.000000e
11	stress2	~~	stress2	0	0.233499998	0.01801124	12.9641256	0.000000e
12	stress3	~~	stress3	0	0.140384332	0.01830990	7.6671251	1.754152e
13	resil1	~~	resil1	0	0.145902374	0.01563347	9.3326916	0.000000e
14	resil2	~~	resil2	0	0.237165314	0.01806223	13.1304527	0.000000e
15	resil3	~~	resil3	0	0.164328836	0.01895948	8.6673701	0.000000e
16	dep1	~~	dep1	0	0.165647815	0.01695952	9.7672450	0.000000e
17	dep2	~~	dep2	0	0.232658800	0.01803935	12.8972935	0.000000e
18	dep3	~~	dep3	0	0.173267425	0.02052210	8.4429675	0.000000e
19	stress	~~	stress	0	0.552502279	0.04651493	11.8779567	0.000000e
20	resilience	~~	resilience	0	0.595146428	0.04765206	12.4894168	0.000000e
21	depression	~~	depression	0	1.244309537	0.08955462	13.8944197	0.000000e
22	stress	~~	resilience	0	-0.296230974	0.03244527	-9.1301739	0.000000e
23	stress	~~	depression	0	0.591172397	0.05054565	11.6958112	0.000000e
24	resilience	~~	depression	0	-0.595649001	0.05120134	-11.6334641	0.000000e
25	stress1	~1		0	0.051164260	0.03827007	1.3369264	1.812467e
26	stress2	~1		0	-0.007807425	0.03660251	-0.2133030	8.310906e
27	stress3	~1		0	0.040657971	0.04220623	0.9633169	3.353885e
28	resil1	~1		0	-0.024374481	0.03849802	-0.6331360	5.266449e
29	resil2	~1		0	-0.011269369	0.03658151	-0.3080619	7.580353e
30	resil3	~1		0	-0.035980507	0.04280506	-0.8405667	4.005907e
31	dep1	~1		0	0.037102437	0.05310287	0.6986899	4.847459e
32	dep2	~1		0	0.047620784	0.04769410	0.9984627	3.180550e
33	dep3	~1		0	0.069974908	0.06016034	1.1631401	2.447727e

This checks whether the measurement structure is reasonable before adding structural paths.

A good workflow is:

```
Step 1: Check the measurement model using CFA
Step 2: Check reliability and factor loadings
Step 3: Fit the full SEM model
Step 4: Interpret direct, indirect, and total effects
Step 5: Report model fit and standardized estimates
```

1.15 16. SEM versus regression

Regression and SEM are related, but SEM is more general.

Feature	Regression	SEM
Observed variables	Yes	Yes
Latent variables	No	Yes
Multiple outcomes	Limited	Yes
Measurement error	Usually ignored	Explicitly modeled
Mediation	Possible	Natural framework
Model fit	Usually not covariance-based	Central part of SEM
Theory testing	Limited	Strong

Regression is often enough when the research question is simple and variables are directly observed.

SEM is useful when:

- constructs are latent,
- measurement error matters,
- several equations are needed,
- mediation is central,
- theory structure is important.

1.16 17. Assumptions of SEM

SEM has assumptions that should be considered carefully.

1.16.1 17.1 Correct model specification

The model should be theoretically justified. SEM can test whether a model is consistent with data, but it cannot prove that the model is true.

A good-fitting model is not necessarily the only good-fitting model.

1.16.2 17.2 Adequate sample size

SEM usually requires moderate to large sample sizes.

There is no single universal rule, but small samples can lead to unstable estimates, convergence problems, or unreliable fit indices.

1.16.3 17.3 Multivariate normality

Classical SEM often assumes multivariate normality, especially with maximum likelihood estimation.

If variables are non-normal, robust estimators can be used.

Example:

```
fit_robust <- sem(
  model_sem,
  data = data_sem,
  estimator = "MLR",
  meanstructure = TRUE
)

summary(
  fit_robust,
  fit.measures = TRUE,
  standardized = TRUE
)
```

$header

$lavaan.version

'0.7-2'

$sam.approach

FALSE

$optim.method

'nlminb'

$optim.iterations

34

$optim.converged

TRUE

$optim

$estimator

'ML'

$estimator.args

$optim.method

'nlminb'

$npar

30

$eq.constraints

FALSE

$nrow.ceq.jac

0

$nrow.cin.jac

0

$nrow.con.jac

0

$con.jac.rank

0

$data

$ngroups

1

$nobs

500

$test

$standard

$test

'standard'

$stat

39.5933180430763

$stat.group

39.5933180430763

$df

24

$refdistr

'chisq'

$pvalue

0.023638187618621

$yuan.bentler.mplus

$test

'yuan.bentler.mplus'

$stat

39.1136150602653

$stat.group

39.1136150602653

$df

24

$pvalue

0.0265687152629294

$scaling.factor

1.01226434790218

$scaling.factor.h1

1.00598789331404

$scaling.factor.h0

1.00096672964353

$label

'Yuan-Bentler correction (Mplus variant)'

$trace.UGamma

24.2943443496523

$trace.UGamma2

<NA>

$scaled.test.stat

39.5933180430763

$scaled.test

'standard'

$fit

npar: 30 **fmin:** 0.0395933180430763 **chisq:** 39.5933180430763 **df:** 24 **pvalue:** 0.023638187618621 **chisq.scaled:** 39.1136150602653 **df.scaled:** 24 **pvalue.scaled:** 0.0265687152629294 **chisq.scaling.factor:** 1.01226434790218 **baseline.chisq:** 4010.69045035666 **baseline.df:** 36 **baseline.pvalue:** 0 **baseline.chisq.scaled:** 3976.21261425848 **baseline.df.scaled:** 36 **baseline.pvalue.scaled:** 0 **baseline.chisq.scaling.factor:** 1.00867102427434 **cfi:** 0.996076847181476 **tli:** 0.994115270772214 **cfi.scaled:** 0.996164264079158 **tli.scaled:** 0.994246396118737 **cfi.robust:** 0.99615059952433 **tli.robust:** 0.994225899286495 **logl:** -4261.75022147873 **unrestricted.logl:** -4241.95356245719 **aic:** 8583.50044295747 **bic:** 8709.93868591013 **ntotal:** 500 **bic2:** 8614.71683163778 **scaling.factor.h1:** 1.00598789331404 **scaling.factor.h0:** 1.00096672964353 **rmsea:** 0.0360477900883863 **rmsea.ci.lower:** 0.0133206537558792 **rmsea.ci.upper:** 0.0554785467010791 **rmsea.ci.level:** 0.9 **rmsea.pvalue:** 0.872609329442447 **rmsea.close.h0:** 0.05 **rmsea.notclose.pvalue:** 2.92071859358351e-05 **rmsea.notclose.h0:** 0.08 **rmsea.scaled:** 0.0354889831030529 **rmsea.ci.lower.scaled:** 0.0124583878620228 **rmsea.ci.upper.scaled:** 0.0548896214404359 **rmsea.pvalue.scaled:** 0.88324809904409 **rmsea.notclose.pvalue.scaled:** 2.18002882029145e-05 **rmsea.robust:** 0.0357059445254148 **rmsea.ci.lower.robust:** 0.0123349424481562 **rmsea.ci.upper.robust:** 0.0553426342690385 **rmsea.pvalue.robust:** 0.876068710519012 **rmsea.notclose.pvalue.robust:** 2.98978644814493e-05 **srmr:** 0.0195584346342506 **gfi:** 0.994230146309336 **gfi.ci.lower:** 0.985281597136529 **gfi.ci.upper:** 0.999811174122434 **gfi.ci.level:** 0.9 **gfi.robust:**

0.994359477575713 **gfi.ci.lower.robust:** 0.985362374926487

gfi.ci.upper.robust: 0.999942565997681

$pe

A data.frame: 36 × 11

	lhs	op	rhs	label	exo	est	se	z	p
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<
1	stress	=~	stress1		0	1.000000000	0.00000000	NA	N
2	stress	=~	stress2		0	0.888712566	0.03942880	22.5396804	0.
3	stress	=~	stress3		0	1.165333138	0.04482736	25.9960266	0.
4	resilience	=~	resil1		0	1.000000000	0.00000000	NA	N
5	resilience	=~	resil2		0	0.851920295	0.03803463	22.3985404	0.
6	resilience	=~	resil3		0	1.123935553	0.04081111	27.5399390	0.
7	depression	=~	dep1		0	1.000000000	0.00000000	NA	N
8	depression	=~	dep2		0	0.852686146	0.02411456	35.3598025	0.
9	depression	=~	dep3		0	1.146769075	0.02812917	40.7679698	0.
10	resilience	~	stress	a	0	-0.536162652	0.04762429	-11.2581761	0.
11	depression	~	resilience	b	0	-0.638717896	0.05328262	-11.9873592	0.
12	depression	~	stress	c	0	0.727534488	0.05931887	12.2648068	0.
13	stress1	~~	stress1		0	0.179796776	0.01502218	11.9687500	0.
14	stress2	~~	stress2		0	0.233500062	0.01873282	12.4647549	0.
15	stress3	~~	stress3		0	0.140384235	0.01809214	7.7594060	8.
16	resil1	~~	resil1		0	0.145902412	0.01518287	9.6096700	0.
17	resil2	~~	resil2		0	0.237165208	0.01635885	14.4976666	0.
18	resil3	~~	resil3		0	0.164328886	0.02044476	8.0377007	8.
19	dep1	~~	dep1		0	0.165647777	0.01833613	9.0339534	0.
20	dep2	~~	dep2		0	0.232658848	0.01754117	13.2635869	0.
21	dep3	~~	dep3		0	0.173267305	0.02076417	8.3445350	0.
22	stress	~~	stress		0	0.552502385	0.04624425	11.9474836	0.
23	resilience	~~	resilience		0	0.436318467	0.03737519	11.6740140	0.
24	depression	~~	depression		0	0.433759482	0.03861860	11.2318795	0.

	lhs	op	rhs	label	exo	est	se	z	p
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<
25	stress1	~1			0	0.051164260	0.03827007	1.3369262	1.
26	stress2	~1			0	-0.007807425	0.03660251	-0.2133030	8.
27	stress3	~1			0	0.040657971	0.04220624	0.9633167	3.
28	resil1	~1			0	-0.024374481	0.03849803	-0.6331359	5.
29	resil2	~1			0	-0.011269369	0.03658152	-0.3080618	7.
30	resil3	~1			0	-0.035980507	0.04280506	-0.8405667	4.
31	dep1	~1			0	0.037102437	0.05310288	0.6986898	4.
32	dep2	~1			0	0.047620784	0.04769411	0.9984627	3.
33	dep3	~1			0	0.069974908	0.06016035	1.1631401	2.
37	indirect	:=	a*b	indirect	0	0.342456681	0.04001743	8.5576878	0.
38	direct	:=	c	direct	0	0.727534488	0.05931887	12.2648068	0.
39	total	:=	c+(a*b)	total	0	1.069991169	0.06527865	16.3911355	0.

`MLR` provides robust standard errors and a robust test statistic.

1.16.4 17.4 Missing data

SEM can handle missing data using full information maximum likelihood.

```
fit_missing <- sem(
  model_sem,
  data = data_sem,
  missing = "fiml",
  meanstructure = TRUE
)
```

This is often preferable to simply deleting all rows with missing values, assuming the missing data mechanism is appropriate.

1.17 18. Modification indices

Modification indices suggest possible model changes that may improve fit.

```
modindices(
  fit_sem,
  sort = TRUE,
```

```
  minimum.value = 10
)
```

A lavaan.data.frame: 2 × 8

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
66	stress2	~~	stress3	15.32568	-0.08218576	-0.08218576	-0.4539352	-0.4539352
52	depression	=~	stress1	15.31238	-0.13934818	-0.15544099	-0.1816441	-0.1816441

However, modification indices should be used cautiously.

You should not blindly add paths just because they improve model fit. Every added path should have theoretical justification.

Otherwise, the model becomes data-driven and may not replicate.

1.18 19. Reporting SEM results

A clear SEM report should include:

1. The theoretical model
2. Description of observed indicators
3. Estimator used
4. Sample size
5. Model fit indices
6. Standardized factor loadings
7. Structural path coefficients
8. Direct, indirect, and total effects if mediation is tested
9. Any model modifications
10. Limitations

Example reporting paragraph:

> We fitted a structural equation model to examine whether resilience mediated the association between stress and depression. Stress, resilience, and depression were modeled as latent variables, each measured by three observed indicators. The model showed acceptable fit according to CFI, TLI, RMSEA, and SRMR. Standardized path estimates indicated that higher

> stress was associated with lower resilience, and higher resilience was associated with lower depression. The indirect effect from stress to depression through resilience was estimated using bootstrapped confidence intervals.

1.19 20. Common mistakes in SEM

1.19.1 Mistake 1: Treating SEM as causal proof

SEM can represent causal hypotheses, but it does not prove causality by itself. Causal interpretation depends on design, theory, temporal ordering, confounding control, and assumptions.

1.19.2 Mistake 2: Ignoring the measurement model

If the latent variables are poorly measured, the structural paths may not be meaningful.

1.19.3 Mistake 3: Chasing fit indices

Good fit indices do not guarantee that the model is scientifically meaningful.

1.19.4 Mistake 4: Overfitting with modification indices

Adding many paths based only on modification indices can make the model fit the current dataset but fail in future datasets.

1.19.5 Mistake 5: Reporting only p-values

SEM results should include effect sizes, standardized estimates, confidence intervals, and model fit indices.

```
## 21. Full R code in one block


# ============================================================
# Full SEM Tutorial Script
# ============================================================

library(lavaan)
library(semPlot)
library(MASS)

set.seed(123)
```

```
n <- 500

stress_latent <- rnorm(n, 0, 1)

resilience_latent <- -0.5 * stress_latent + rnorm(n, 0, 0.8)

depression_latent <- 0.6 * stress_latent -
  0.7 * resilience_latent +
  rnorm(n, 0, 0.8)

data_sem <- data.frame(
  stress1 = 0.8 * stress_latent + rnorm(n, 0, 0.4),
  stress2 = 0.7 * stress_latent + rnorm(n, 0, 0.5),
  stress3 = 0.9 * stress_latent + rnorm(n, 0, 0.4),

  resil1 = 0.8 * resilience_latent + rnorm(n, 0, 0.4),
  resil2 = 0.7 * resilience_latent + rnorm(n, 0, 0.5),
  resil3 = 0.9 * resilience_latent + rnorm(n, 0, 0.4),

  dep1 = 0.8 * depression_latent + rnorm(n, 0, 0.4),
  dep2 = 0.7 * depression_latent + rnorm(n, 0, 0.5),
  dep3 = 0.9 * depression_latent + rnorm(n, 0, 0.4)
)

model_sem <- '

  # Measurement model
  stress =~ stress1 + stress2 + stress3
  resilience =~ resil1 + resil2 + resil3
  depression =~ dep1 + dep2 + dep3

  # Structural model
  resilience ~ a * stress
  depression ~ b * resilience + c * stress

  # Effects
  indirect := a * b
  direct := c
  total := c + (a * b)

'

fit_sem <- sem(
  model_sem,
  data = data_sem,
  meanstructure = TRUE
)
```

```
summary(
  fit_sem,
  fit.measures = TRUE,
  standardized = TRUE,
  rsquare = TRUE
)

fitMeasures(
  fit_sem,
  c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "srmr")
)

parameterEstimates(
  fit_sem,
  standardized = TRUE
)

semPaths(
  fit_sem,
  what = "std",
  layout = "tree",
  edge.label.cex = 1.0,
  sizeMan = 6,
  sizeLat = 8,
  residuals = FALSE,
  intercepts = FALSE,
  nCharNodes = 0
)

fit_sem_boot <- sem(
  model_sem,
  data = data_sem,
  se = "bootstrap",
  bootstrap = 1000,
  meanstructure = TRUE
)

parameterEstimates(
  fit_sem_boot,
  boot.ci.type = "perc",
  standardized = TRUE
)
```

$header

$lavaan.version

'0.7-2'

$sam.approach

FALSE

$optim.method

'nlminb'

$optim.iterations

34

$optim.converged

TRUE

$optim

$estimator

'ML'

$estimator.args

$optim.method

'nlminb'

$npar

30

$eq.constraints

FALSE

$nrow.ceq.jac

0

$nrow.cin.jac

0

$nrow.con.jac

0

$con.jac.rank

0

$data

$ngroups

1

$nobs

500

$test

$standard =

$test

'standard'

$stat

39.5933180430763

$stat.group

39.5933180430763

$df

24

$refdistr

'chisq'

$pvalue

0.023638187618621

$fit

npar: 30 **fmin:** 0.0395933180430763 **chisq:** 39.5933180430763 **df:** 24 **pvalue:** 0.023638187618621 **baseline.chisq:** 4010.69045035666 **baseline.df:** 36 **baseline.pvalue:** 0 **cfi:** 0.996076847181476 **tli:** 0.994115270772214 **logl:** -4261.75022147873 **unrestricted.logl:** -4241.95356245719 **aic:** 8583.50044295747 **bic:** 8709.93868591013 **ntotal:** 500 **bic2:** 8614.71683163778 **rmsea:** 0.0360477900883863 **rmsea.ci.lower:** 0.0133206537558792 **rmsea.ci.upper:** 0.0554785467010791 **rmsea.ci.level:** 0.9 **rmsea.pvalue:** 0.872609329442447 **rmsea.close.h0:** 0.05 **rmsea.notclose.pvalue:** 2.92071859358351e-05 **rmsea.notclose.h0:** 0.08 **srmr:** 0.0195584346342506 **gfi:** 0.994230146309336 **gfi.ci.lower:** 0.985281597136529 **gfi.ci.upper:** 0.999811174122434 **gfi.ci.level:** 0.9

$pe

A data.frame: 47 × 11

	lhs	op	rhs	label	exo	est	se	z	pv
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<
1	stress	=~	stress1		0	1.000000000	0.00000000	NA	N
2	stress	=~	stress2		0	0.888712566	0.03965218	22.4127048	0.
3	stress	=~	stress3		0	1.165333138	0.04332610	26.8967905	0.
4	resilience	=~	resil1		0	1.000000000	0.00000000	NA	N
5	resilience	=~	resil2		0	0.851920295	0.03685346	23.1164239	0.
6	resilience	=~	resil3		0	1.123935553	0.04011596	28.0171692	0.
7	depression	=~	dep1		0	1.000000000	0.00000000	NA	N
8	depression	=~	dep2		0	0.852686146	0.02485828	34.3018961	0.
9	depression	=~	dep3		0	1.146769075	0.02726355	42.0623568	0.
10	resilience	~	stress	a	0	-0.536162652	0.04798759	-11.1729444	0.
11	depression	~	resilience	b	0	-0.638717896	0.05641726	-11.3213206	0.
12	depression	~	stress	c	0	0.727534488	0.05968082	12.1904241	0.
13	stress1	~~	stress1		0	0.179796776	0.01664951	10.7989246	0.
14	stress2	~~	stress2		0	0.233500062	0.01801124	12.9641268	0.
15	stress3	~~	stress3		0	0.140384235	0.01830990	7.6671204	1.
16	resil1	~~	resil1		0	0.145902412	0.01563347	9.3326929	0.
17	resil2	~~	resil2		0	0.237165208	0.01806223	13.1304497	0.
18	resil3	~~	resil3		0	0.164328886	0.01895948	8.6673739	0.
19	dep1	~~	dep1		0	0.165647777	0.01695952	9.7672442	0.
20	dep2	~~	dep2		0	0.232658848	0.01803935	12.8972957	0.
21	dep3	~~	dep3		0	0.173267305	0.02052209	8.4429653	0.
22	stress	~~	stress		0	0.552502385	0.04651494	11.8779564	0.
23	resilience	~~	resilience		0	0.436318467	0.03668041	11.8951349	0.
24	depression	~~	depression		0	0.433759482	0.03669715	11.8199764	0.
25	stress1	~1			0	0.051164260	0.03827007	1.3369262	1.
26	stress2	~1			0	-0.007807425	0.03660251	-0.2133030	8.
27	stress3	~1			0	0.040657971	0.04220624	0.9633167	3.
28	resil1	~1			0	-0.024374481	0.03849802	-0.6331359	5.

	lhs	op	rhs	label	exo	est	se	z	p
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<
29	resil2	~1			0	-0.011269369	0.03658152	-0.3080618	7.
30	resil3	~1			0	-0.035980507	0.04280506	-0.8405667	4.
31	dep1	~1			0	0.037102437	0.05310287	0.6986898	4.
32	dep2	~1			0	0.047620784	0.04769410	0.9984627	3.
33	dep3	~1			0	0.069974908	0.06016034	1.1631401	2.
37	indirect	:=	a*b	indirect	0	0.342456681	0.04061961	8.4308214	0.
38	direct	:=	c	direct	0	0.727534488	0.05968082	12.1904241	0.
39	total	:=	c+(a*b)	total	0	1.069991169	0.06169700	17.3426763	0.
40	stress1	r2	stress1		0	0.754476332	NA	NA	N
41	stress2	r2	stress2		0	0.651425844	NA	NA	N
42	stress3	r2	stress3		0	0.842385896	NA	NA	N
43	resil1	r2	resil1		0	0.803113666	NA	NA	N
44	resil2	r2	resil2		0	0.645547872	NA	NA	N
45	resil3	r2	resil3		0	0.820628388	NA	NA	N
46	dep1	r2	dep1		0	0.882515768	NA	NA	N
47	dep2	r2	dep2		0	0.795440236	NA	NA	N
48	dep3	r2	dep3		0	0.904252809	NA	NA	N
49	resilience	r2	resilience		0	0.266872214	NA	NA	N
50	depression	r2	depression		0	0.651405541	NA	NA	N

chisq: 39.5933180430763 **df:** 24 **pvalue:** 0.023638187618621 **cfi:** 0.996076847181476 **tli:** 0.994115270772214 **rmsea:** 0.0360477900883863 **srmr:** 0.0195584346342506

A lavaan.data.frame: 39 × 12

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
stress	=~	stress1		1.000000000	0.00000000	NA	NA
stress	=~	stress2		0.888712566	0.03965218	22.4127048	0.000000e+
stress	=~	stress3		1.165333138	0.04332610	26.8967905	0.000000e+

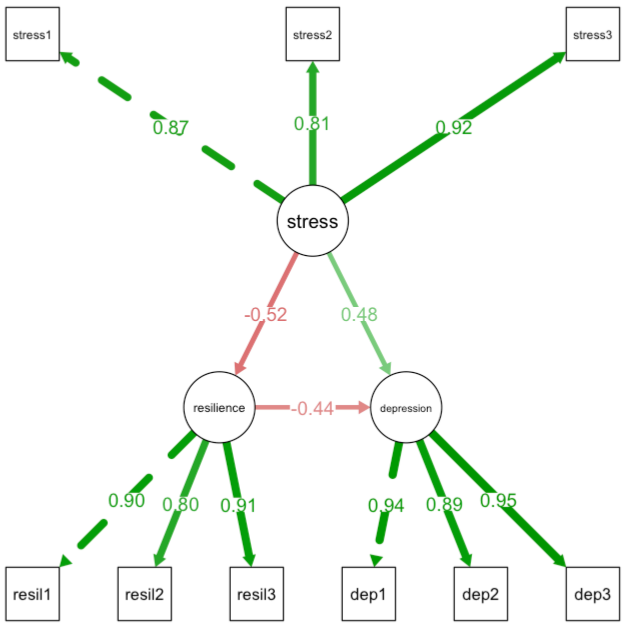
lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
resilience	=~	resil1		1.000000000	0.00000000	NA	NA
resilience	=~	resil2		0.851920295	0.03685346	23.1164239	0.000000e+
resilience	=~	resil3		1.123935553	0.04011596	28.0171692	0.000000e+
depression	=~	dep1		1.000000000	0.00000000	NA	NA
depression	=~	dep2		0.852686146	0.02485828	34.3018961	0.000000e+
depression	=~	dep3		1.146769075	0.02726355	42.0623568	0.000000e+
resilience	~	stress	a	-0.536162652	0.04798759	-11.1729444	0.000000e+
depression	~	resilience	b	-0.638717896	0.05641726	-11.3213206	0.000000e+
depression	~	stress	c	0.727534488	0.05968082	12.1904241	0.000000e+
stress1	~~	stress1		0.179796776	0.01664951	10.7989246	0.000000e+
stress2	~~	stress2		0.233500062	0.01801124	12.9641268	0.000000e+
stress3	~~	stress3		0.140384235	0.01830990	7.6671204	1.754152e-
resil1	~~	resil1		0.145902412	0.01563347	9.3326929	0.000000e+
resil2	~~	resil2		0.237165208	0.01806223	13.1304497	0.000000e+
resil3	~~	resil3		0.164328886	0.01895948	8.6673739	0.000000e+
dep1	~~	dep1		0.165647777	0.01695952	9.7672442	0.000000e+
dep2	~~	dep2		0.232658848	0.01803935	12.8972957	0.000000e+
dep3	~~	dep3		0.173267305	0.02052209	8.4429653	0.000000e+
stress	~~	stress		0.552502385	0.04651494	11.8779564	0.000000e+
resilience	~~	resilience		0.436318467	0.03668041	11.8951349	0.000000e+
depression	~~	depression		0.433759482	0.03669715	11.8199764	0.000000e+
stress1	~1			0.051164260	0.03827007	1.3369262	1.812467e-
stress2	~1			-0.007807425	0.03660251	-0.2133030	8.310907e-
stress3	~1			0.040657971	0.04220624	0.9633167	3.353886e-
resil1	~1			-0.024374481	0.03849802	-0.6331359	5.266449e-
resil2	~1			-0.011269369	0.03658152	-0.3080618	7.580353e-
resil3	~1			-0.035980507	0.04280506	-0.8405667	4.005907e-
dep1	~1			0.037102437	0.05310287	0.6986898	4.847459e-
dep2	~1			0.047620784	0.04769410	0.9984627	3.180550e-

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
dep3	~1			0.069974908	0.06016034	1.1631401	2.447726e-
stress	~1			0.000000000	0.00000000	NA	NA
resilience	~1			0.000000000	0.00000000	NA	NA
depression	~1			0.000000000	0.00000000	NA	NA
indirect	:=	a*b	indirect	0.342456681	0.04061961	8.4308214	0.000000e+
direct	:=	c	direct	0.727534488	0.05968082	12.1904241	0.000000e+
total	:=	c+(a*b)	total	1.069991169	0.06169700	17.3426763	0.000000e+

A lavaan.data.frame: 39 × 12

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
stress	=~	stress1		1.000000000	0.00000000	NA	NA
stress	=~	stress2		0.888712566	0.03875278	22.9328715	0.000000e+
stress	=~	stress3		1.165333138	0.04330515	26.9098040	0.000000e+
resilience	=~	resil1		1.000000000	0.00000000	NA	NA
resilience	=~	resil2		0.851920295	0.03808626	22.3681820	0.000000e+
resilience	=~	resil3		1.123935553	0.04317346	26.0330200	0.000000e+
depression	=~	dep1		1.000000000	0.00000000	NA	NA
depression	=~	dep2		0.852686146	0.02355667	36.1972280	0.000000e+
depression	=~	dep3		1.146769075	0.02873547	39.9077899	0.000000e+
resilience	~	stress	a	-0.536162652	0.04772853	-11.2335886	0.000000e+
depression	~	resilience	b	-0.638717896	0.05169940	-12.3544541	0.000000e+
depression	~	stress	c	0.727534488	0.05869613	12.3949303	0.000000e+
stress1	~~	stress1		0.179796776	0.01502245	11.9685414	0.000000e+
stress2	~~	stress2		0.233500062	0.01885213	12.3858737	0.000000e+
stress3	~~	stress3		0.140384235	0.01843239	7.6161708	2.620126e-
resil1	~~	resil1		0.145902412	0.01595743	9.1432259	0.000000e+
resil2	~~	resil2		0.237165208	0.01709337	13.8746937	0.000000e+
resil3	~~	resil3		0.164328886	0.02084598	7.8830019	3.108624e-

lhs	op	rhs	label	est	se	z	pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
dep1	~~	dep1		0.165647777	0.01906716	8.6875974	0.000000e+
dep2	~~	dep2		0.232658848	0.01764236	13.1875122	0.000000e+
dep3	~~	dep3		0.173267305	0.02048533	8.4581171	0.000000e+
stress	~~	stress		0.552502385	0.04695879	11.7656873	0.000000e+
resilience	~~	resilience		0.436318467	0.03765419	11.5875151	0.000000e+
depression	~~	depression		0.433759482	0.03872977	11.1996388	0.000000e+
stress1	~1			0.051164260	0.03771417	1.3566322	1.748981e-(
stress2	~1			-0.007807425	0.03691312	-0.2115081	8.324908e-(
stress3	~1			0.040657971	0.04279942	0.9499654	3.421298e-(
resil1	~1			-0.024374481	0.03766567	-0.6471272	5.175496e-(
resil2	~1			-0.011269369	0.03674697	-0.3066748	7.590909e-(
resil3	~1			-0.035980507	0.04224272	-0.8517564	3.943493e-(
dep1	~1			0.037102437	0.05255557	0.7059658	4.802094e-(
dep2	~1			0.047620784	0.04743160	1.0039886	3.153841e-(
dep3	~1			0.069974908	0.05921142	1.1817807	2.372927e-(
stress	~1			0.000000000	0.00000000	NA	NA
resilience	~1			0.000000000	0.00000000	NA	NA
depression	~1			0.000000000	0.00000000	NA	NA
indirect	:=	a*b	indirect	0.342456681	0.03929570	8.7148636	0.000000e+
direct	:=	c	direct	0.727534488	0.05869613	12.3949303	0.000000e+
total	:=	c+(a*b)	total	1.069991169	0.06601319	16.2087480	0.000000e+



1.20 22. Final takeaway

Structural Equation Modeling is a powerful method for studying complex relationships among observed and latent variables. It is especially useful when a research question involves measurement error, latent constructs, mediation, and multiple pathways.

A good SEM analysis should not start from software. It should start from theory.

The main question is not only:

> Which paths are significant?

A better question is:

> Does this theoretical model provide a reasonable and interpretable explanation of the observed data?

SEM is strongest when statistical modeling, measurement quality, and domain theory are used together.

1.21 References

Rosseel, Y. (2012). *lavaan: An R Package for Structural Equation Modeling*. Journal of Statistical Software, 48(2), 1–36.

Official lavaan tutorials: https://lavaan.ugent.be/tutorial/

Kline, R. B. *Principles and Practice of Structural Equation Modeling*.